Zihan Nie

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Gender: Female **Date of Birth:** 10 July 2002
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Education

Shandong University

Master of Engineering in Mechanical Engineering **Expected June 2027**
Cumulative GPA (current): 88.2/100

Bachelor of Engineering in Mechanical Design, Manufacturing and Automation **June 2024**
Cumulative GPA: 3.89/4.0 (Top 10%)

Publications and Patents

Published

- **Z Nie**, M Xu, Y Cui, H Wei, W Yi, S Niu, Y Wan, X Wei, W Song. **"Few-shot Medical Anomaly Detection through Centroid Consultation Back and Test-Time Self-Calibration"**. *Pattern Recognition (PR)*, 2026.
- M Xu, **Z Nie**, B Fu, Z Chen, Z Li, H Wei, Y Wan, W Song. **"Beyond Feature Mapping: Dual-Heterogeneous Knowledge Distillation with Mamba for Industrial Anomaly Detection"**. *Expert Systems with Applications (ESWA)*, 2026.
- M Xu[†], L Qu[†], **Z Nie,** F Li, Z Chen, Q Liao, Y Wan, S Niu, R Cong, X Li, W Song. **"Towards reliable medical unsupervised anomaly detection: A benchmark and dataset for PET/CT with cross-modality knowledge distillation network"**. *Information Fusion (IF)*, 2026.
- M Xu, H Wei, **Z Nie**, X Zhou, B Fu, H Yan, Y Wan, W Song. **"Hybrid Dual-Heterogeneous Knowledge Distillation Network for Anomaly Detection in Retinal OCT Images"**. *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2025.
- **Z Nie**, M Xu, Z Wang, X Lu, W Song. **"A Review of Application of Deep Learning in Endoscopic Image Processing".** *Journal of Imaging*, 2024.

Manuscripts Under Review

- **Z Nie** et al. **"ReMoE: Report-Guided Mixture-of-Experts for Multimodal OCT/OCTA Anomaly Detection"**. Under review at *AAAI Conference on Artificial Intelligence (AAAI).*
- **Z Nie** et al. **"CFR-Net: Collaborative Feature Refinement Network for Medical Image Anomaly Detection"**. Under review at *Journal of Biomedical and Health Informatics (JBHI).*

Chinese Invention Patents

- W Song (Advisor), **Z Nie**, Y Wan, X Liang, J Wang, Z Li, K Yang, Y Zhao, Y Lv. **"A Wear Error Compensation System for Wheel Diameter Gauge Positioning Blocks"**. China Patent ZL202410079723.9, granted November 2024.

Academic Service

Reviewer for Pattern Recognition (PR), AAAI, ICME, and BSPC .

Research Experience

Intelligent Robotics and Imaging Systems Engineering Laboratory - Shandong University **July 2023 - Present**

- Advised by: Professors Weiye Song and Yi Wan
- Research:
  - Developed multiple **few-shot** and **unsupervised anomaly detection** frameworks with applications to medical imaging, focusing on **vision-language representation** and score calibration, cross-space consistency-guided **teacher-student distillation**, and **multimodal imaging fusion**.
    (1) Designed a **CLIP-based few-shot anomaly detection framework** with Centroid Consultation Back and Test-Time Self-Calibration to improve representation robustness and decision reliability.
    (2) Proposed a **report-guided multimodal OCT/OCTA anomaly detection framework** that uses clinical report semantics to guide modality-aware fusion of retinal structural and superficial/deep vascular flow information.
    (3) Proposed CFR-Net, an **unsupervised** anomaly detection framework that advances teacher-student distillation through input-side collaborative feature refinement and output-side cross-space consistency.
  - Contributed to high-resolution OCT imaging system development and functional retinal imaging analysis, including visible/near-infrared OCT, endoscopic OCT validation, and retinal oxygen saturation measurement.

Monash University - Research Assistant **July 2026 - Present**

- Research:
  - Built **multimodal predictive modeling pipelines** integrating retinal fundus images, structured clinical variables, and interpretable retinal biomarkers, including A/V ratio and cup-to-disc ratio.
  - Developed image – clinical fusion models for early preeclampsia risk prediction **as early as 14 weeks** of gestation and **progression-oriented myopia risk assessment**, predicting whether existing myopia would further worsen based on current refractive status.

Cardiovascular Imaging Intelligent Analysis Research Group **February 2023 - September 2024**

- Advised by: Professors Jie Tian, Enqing Dong, Zhenchao Tang, and Peng Xue
- Research:
  - Developed an deep learning pipeline for coronary artery tree s**egmentation** from **3D** Coronary CT Angiography (CTA) volumes, incorporating enhanced multi-scale feature extraction and topology-preserving post-processing.
  - Achieved a Dice score of 0.837, demonstrating effectiveness for quantitative 3D cardiovascular image analysis.

Professional Skills

- **Programming Languages:** Python, C
- **Deep Learning Frameworks:** PyTorch
- **Models & Techniques:** Vision-language models, Transformer-based models
- **Image Analysis & Annotation Tools:** DarkLabel, ITK-SNAP, 3D Slicer
- **Simulation & Design Software:** Abaqus, ANSYS, Zemax, SolidWorks

Research Interests

- **Vision-language and multimodal representation learning**
- **World models for embodied intelligence**
- **Robust and generalizable robot learning**

Honors and Awards

- **National Scholarship**, 2026
- **National First Prize,** National Advanced Computing Technology Innovation Competition, 2026
- **National Second Prize,** China International College Students' Innovation Competition, 2024
- **National Second Prize,** 18th National University Students Intelligent Car Race, 2023
- **(Twice) First-class Academic Scholarship**, Shandong University, 2025, 2023
- **Graduate Freshman Scholarship**, Shandong University, 2024
- **Second-class Academic Scholarship**, Shandong University, 2022
- **Third-class Academic Scholarship**, Shandong University, 2021